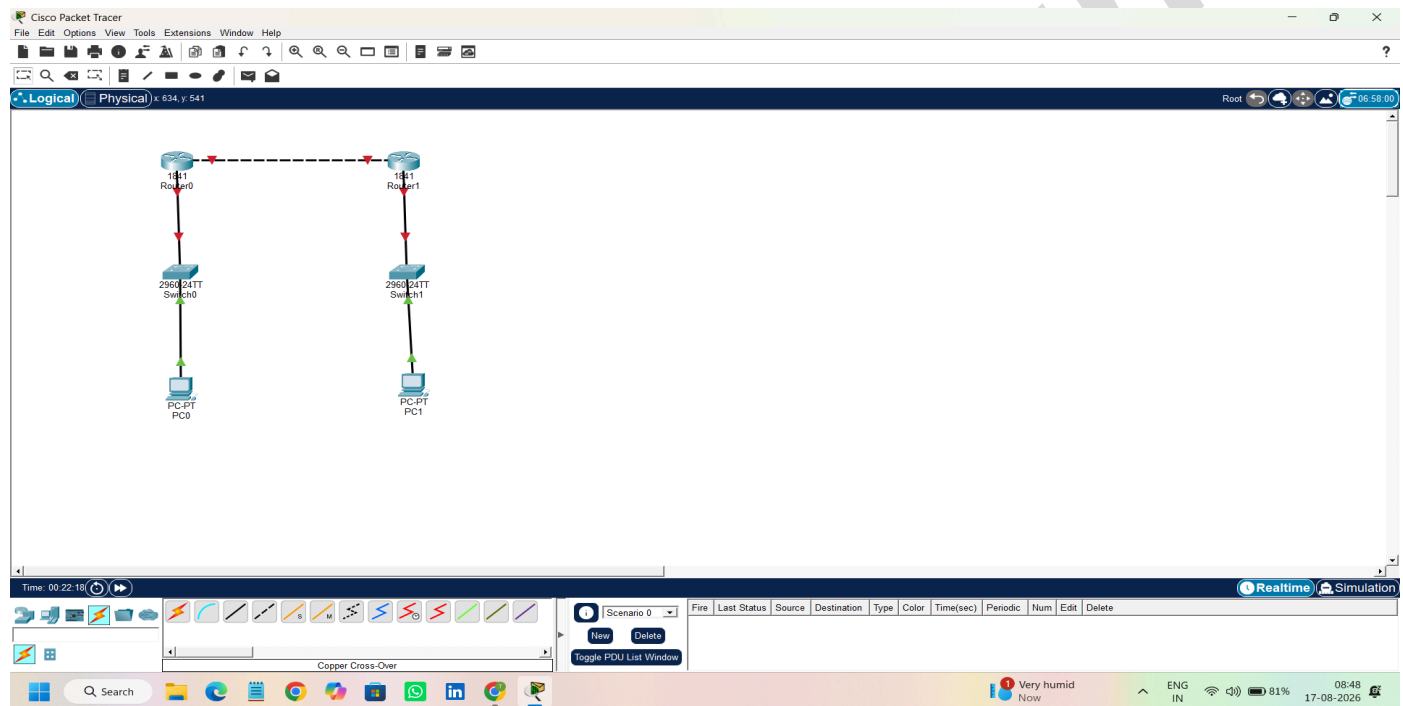
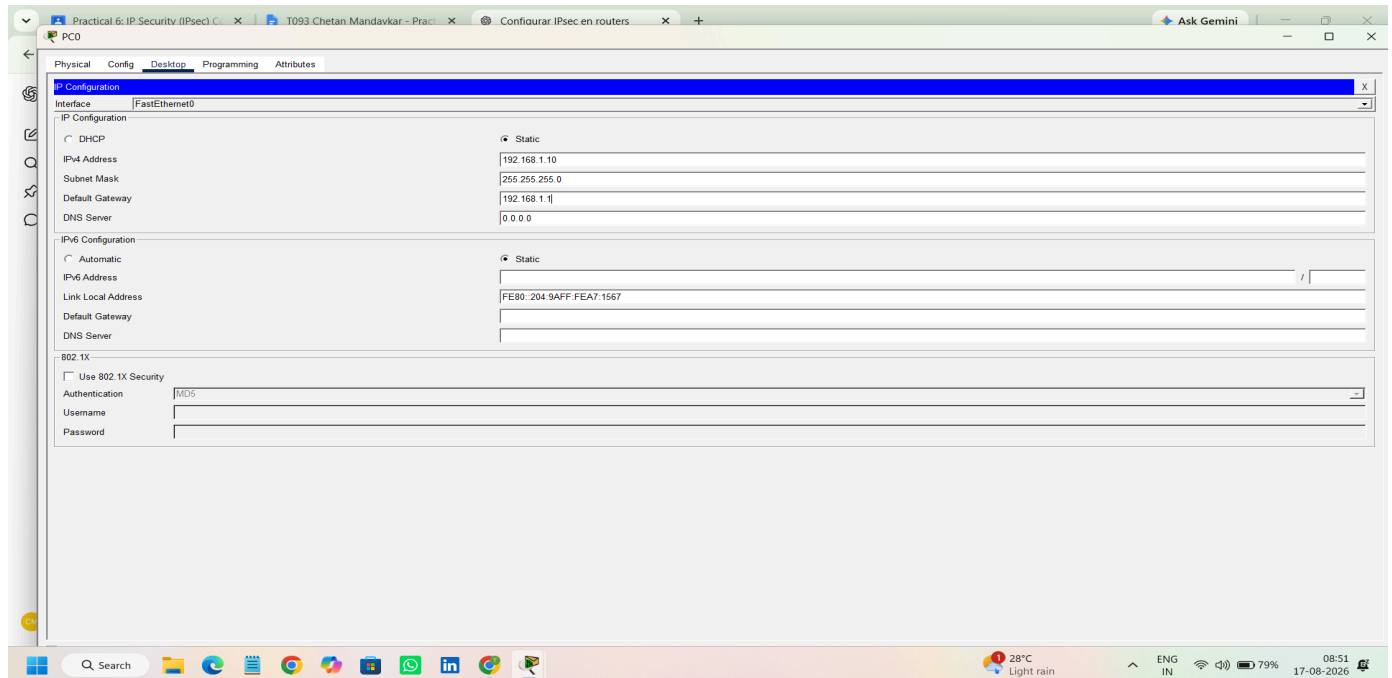


Aim:- Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

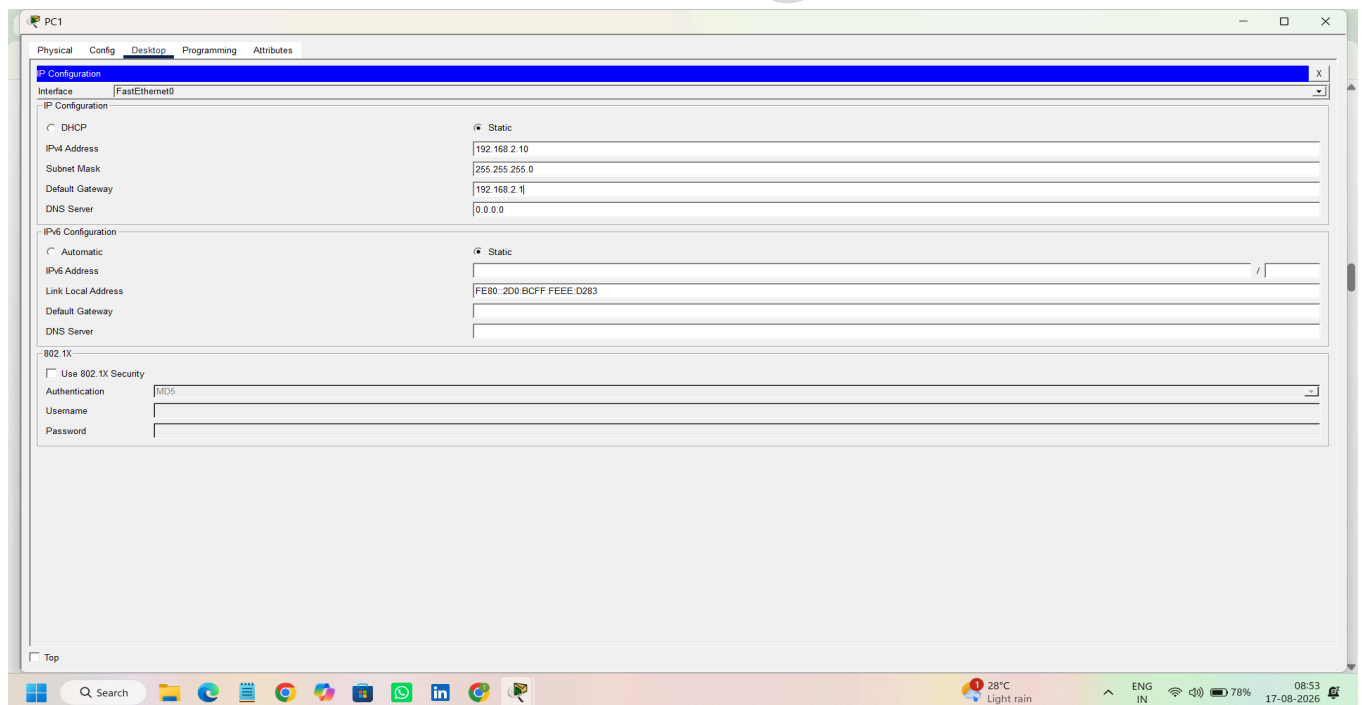
Step 1:- Open Cisco Packet And Add 2 Routers, 2 Switches, 2 PCs And Connect Them To Copper Straight Through And Router Connect With Copper Cross Over



Step 2:- Configure PCs
PC0:-

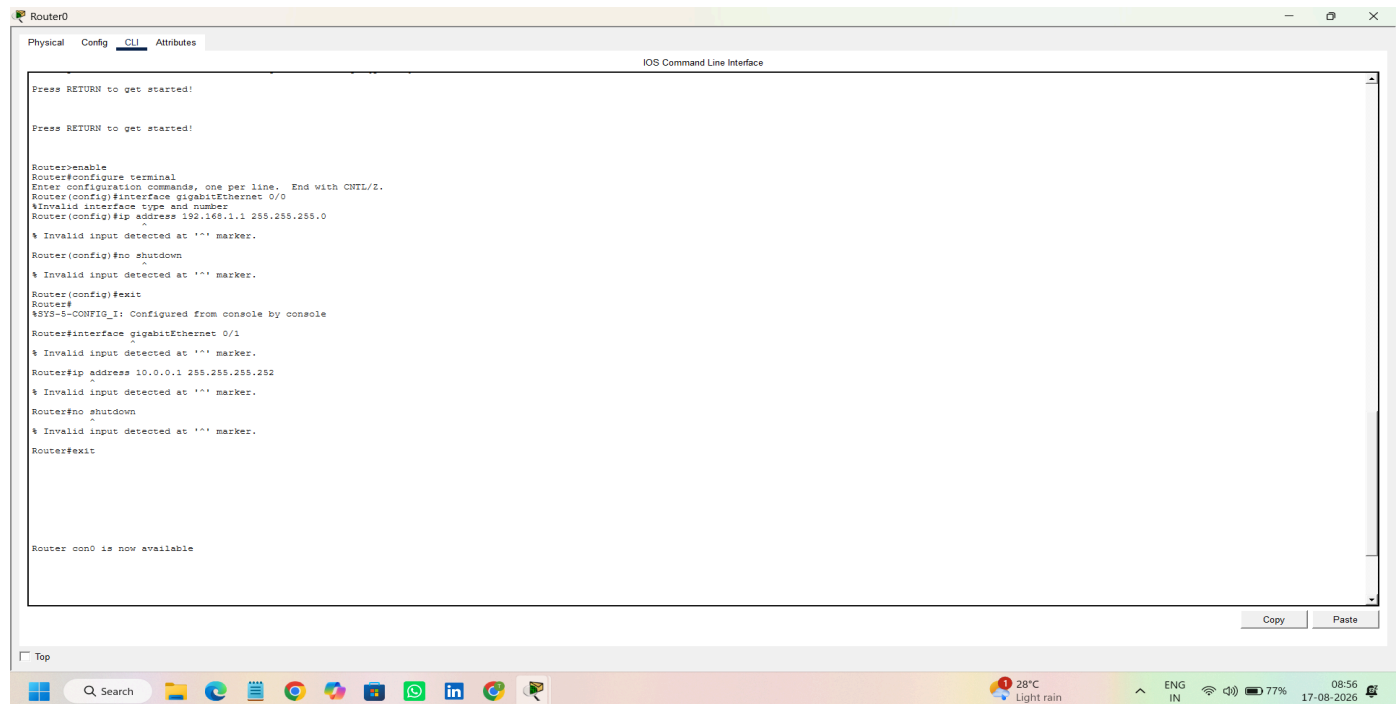


PC1:-



Step 3:- Configure Routers

Router 0:-



The screenshot shows the Router0 CLI interface with the following text:

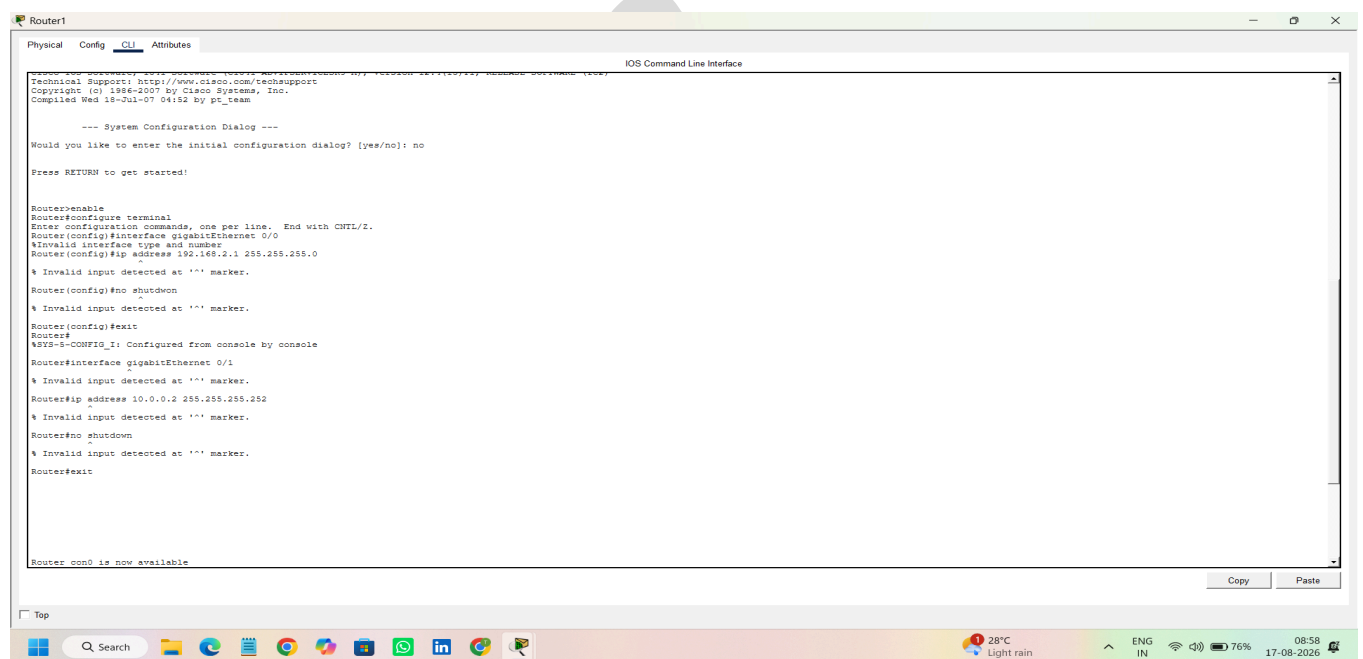
```
Press RETURN to get started!

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
%Invalid interface type and number
Router(config)#ip address 192.168.1.1 255.255.255.0
% Invalid input detected at '^' marker.
Router(config)#no shutdown
% Invalid input detected at '^' marker.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#ip address 10.0.0.1 255.255.255.252
% Invalid input detected at '^' marker.
Router#no shutdown
% Invalid input detected at '^' marker.
Router#exit

Router con0 is now available
```

Router 1:-



The screenshot shows the Router1 CLI interface with the following text:

```
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no

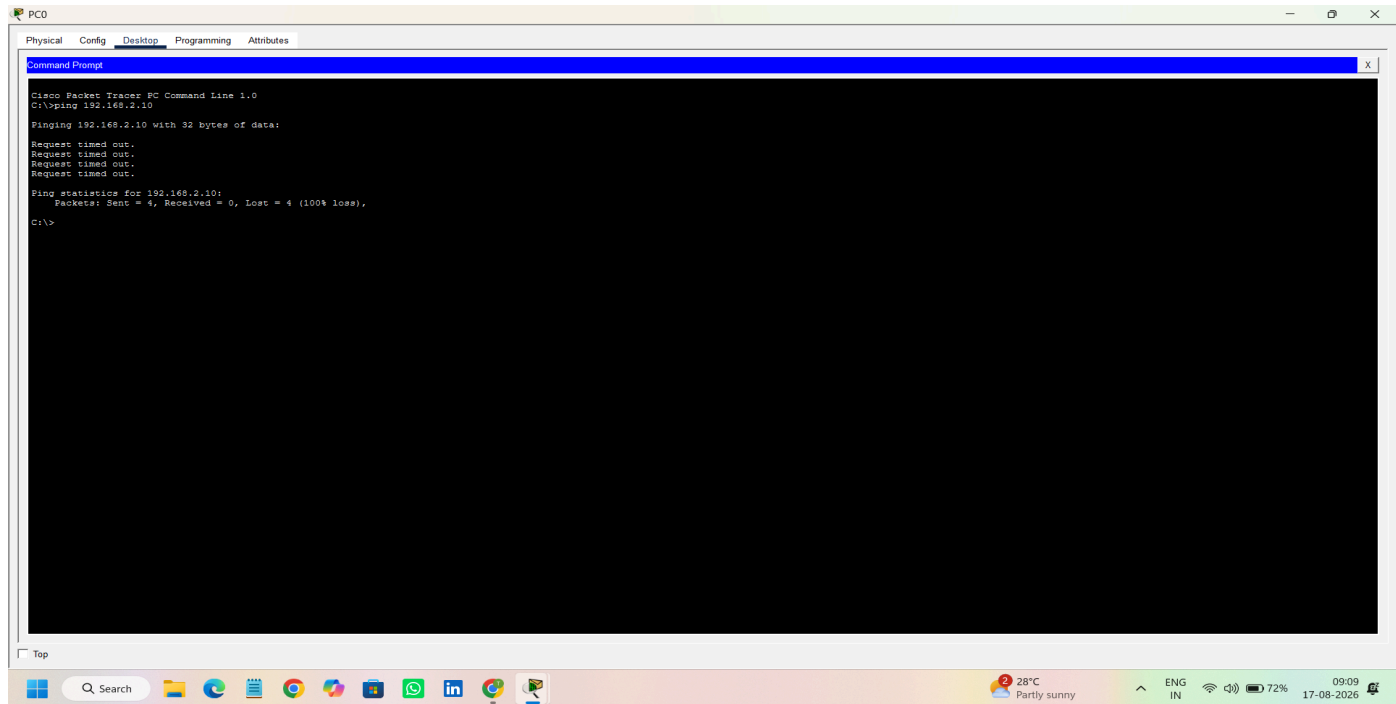
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
%Invalid interface type and number
Router(config)#ip address 192.168.2.1 255.255.255.0
% Invalid input detected at '^' marker.
Router(config)#no shutdown
% Invalid input detected at '^' marker.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#ip address 10.0.0.2 255.255.255.252
% Invalid input detected at '^' marker.
Router#no shutdown
% Invalid input detected at '^' marker.
Router#exit

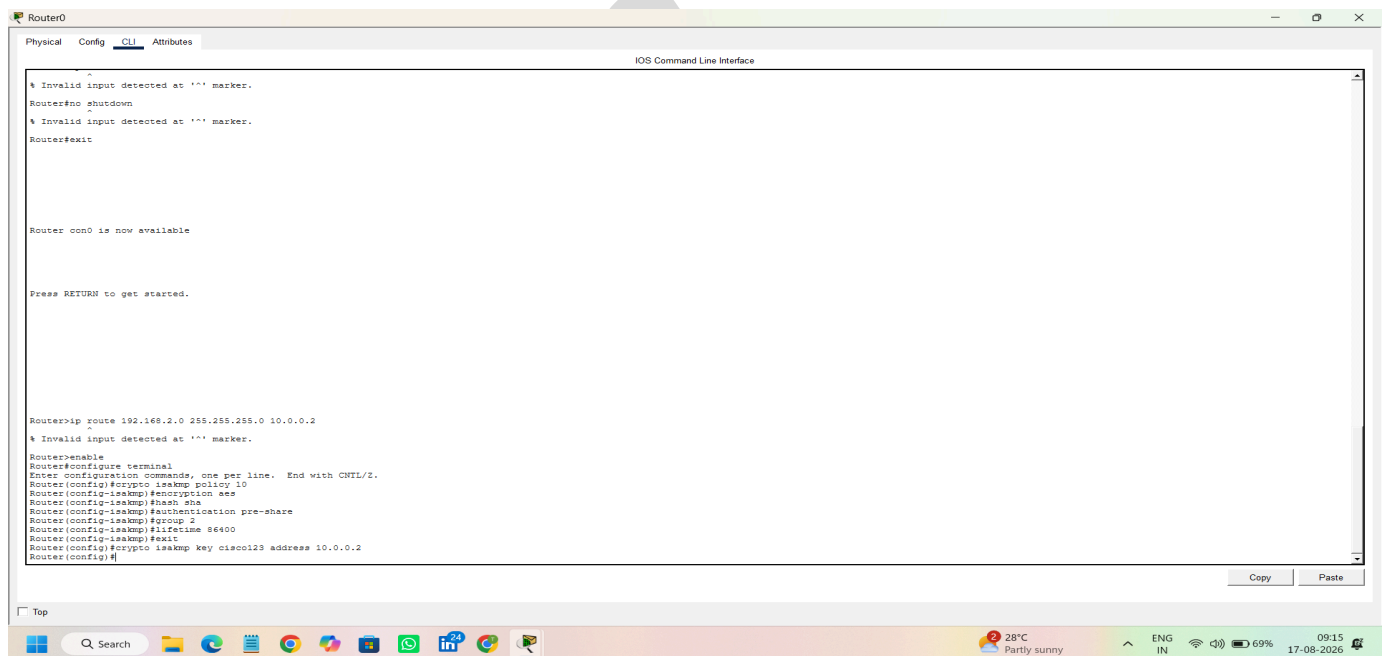
Router con0 is now available
```

Step 4:- Test basic connectivity

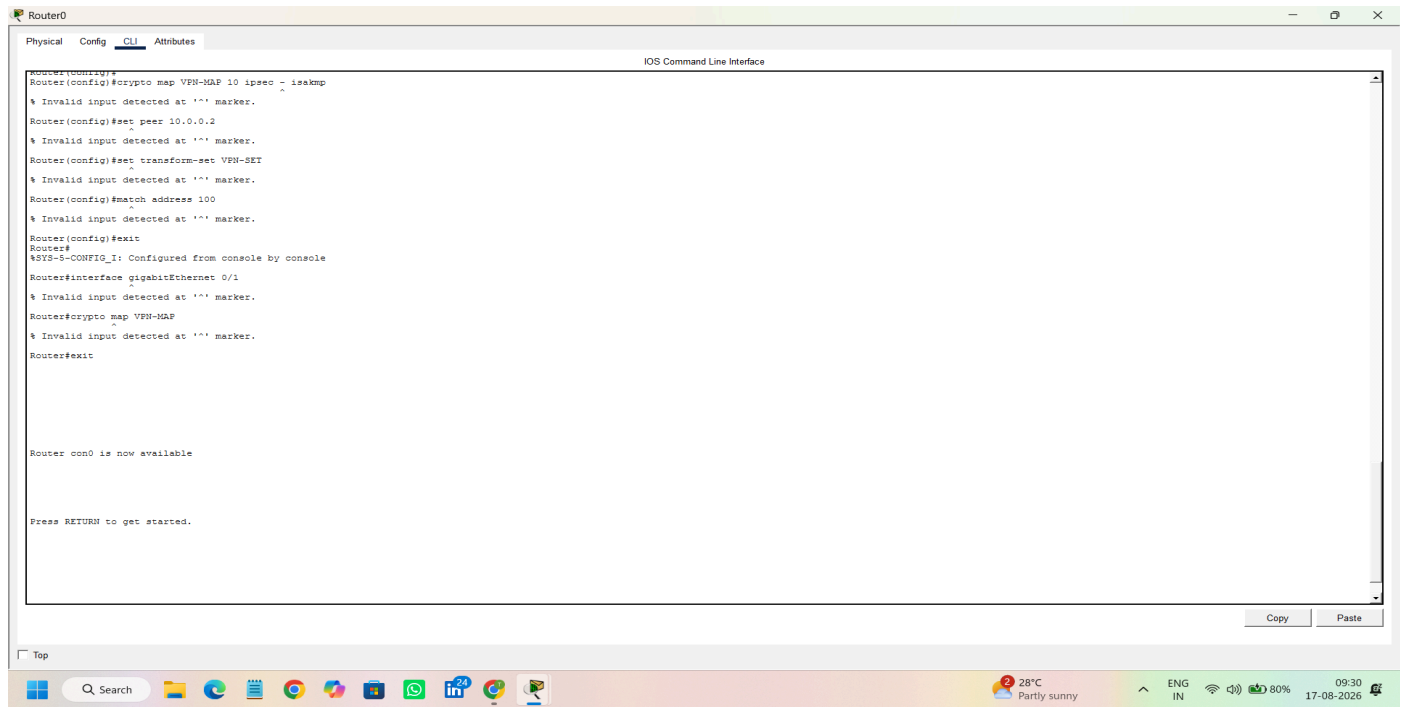
PC0:-



Step 5:- Configure IKE Phase 1 on Router0



Step 10:- Apply Crypto Map to Router0 WAN interface



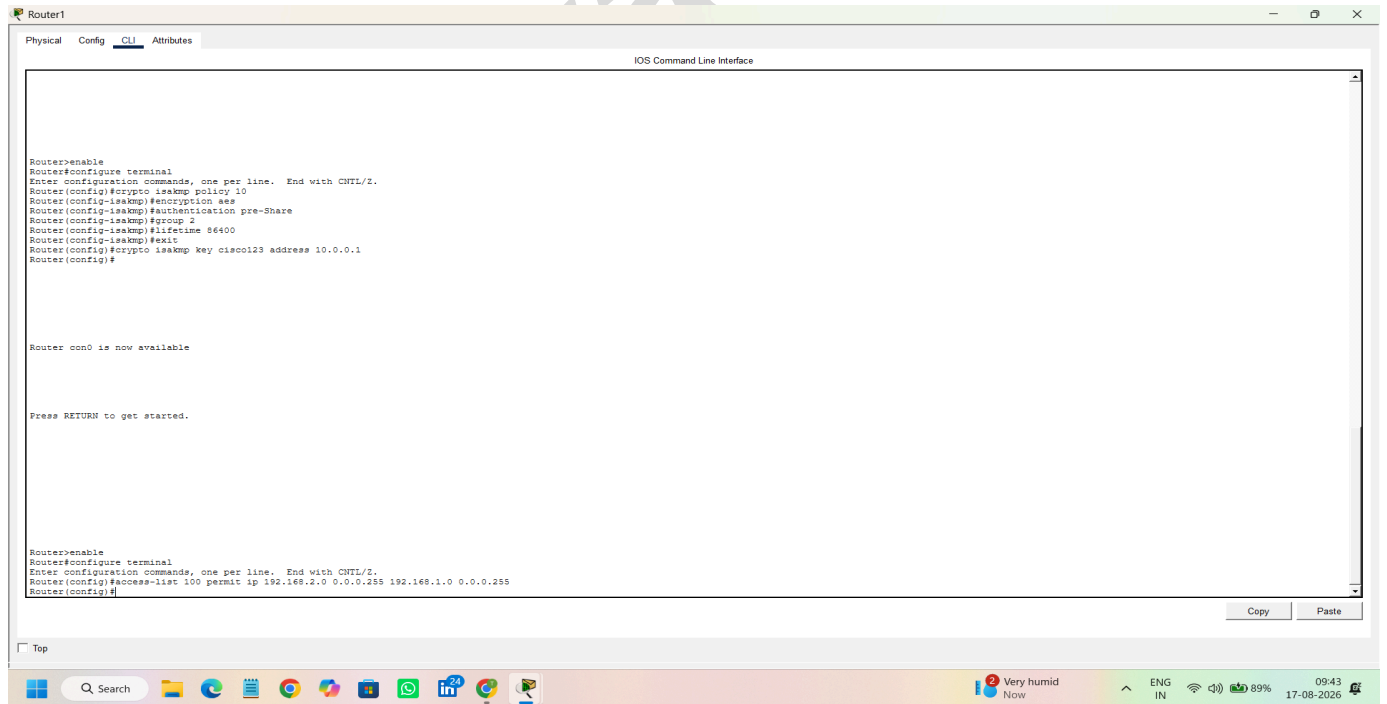
The screenshot shows the CLI of Router0 with the following configuration commands and output:

```
Router>configure
Router(config)#crypto map VPN-MAP 10 ipsec - isakmp
% Invalid input detected at '^' marker.
Router(config)#set peer 10.0.0.2
% Invalid input detected at '^' marker.
Router(config)#set transform-set VPN-SET
% Invalid input detected at '^' marker.
Router(config)#match address 100
% Invalid input detected at '^' marker.
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#crypto map VPN-MAP
% Invalid input detected at '^' marker.
Router#exit

Router oom0 is now available

Press RETURN to get started.
```

Step 11:- Configure IPsec ACL on Router1



The screenshot shows the CLI of Router1 with the following configuration commands and output:

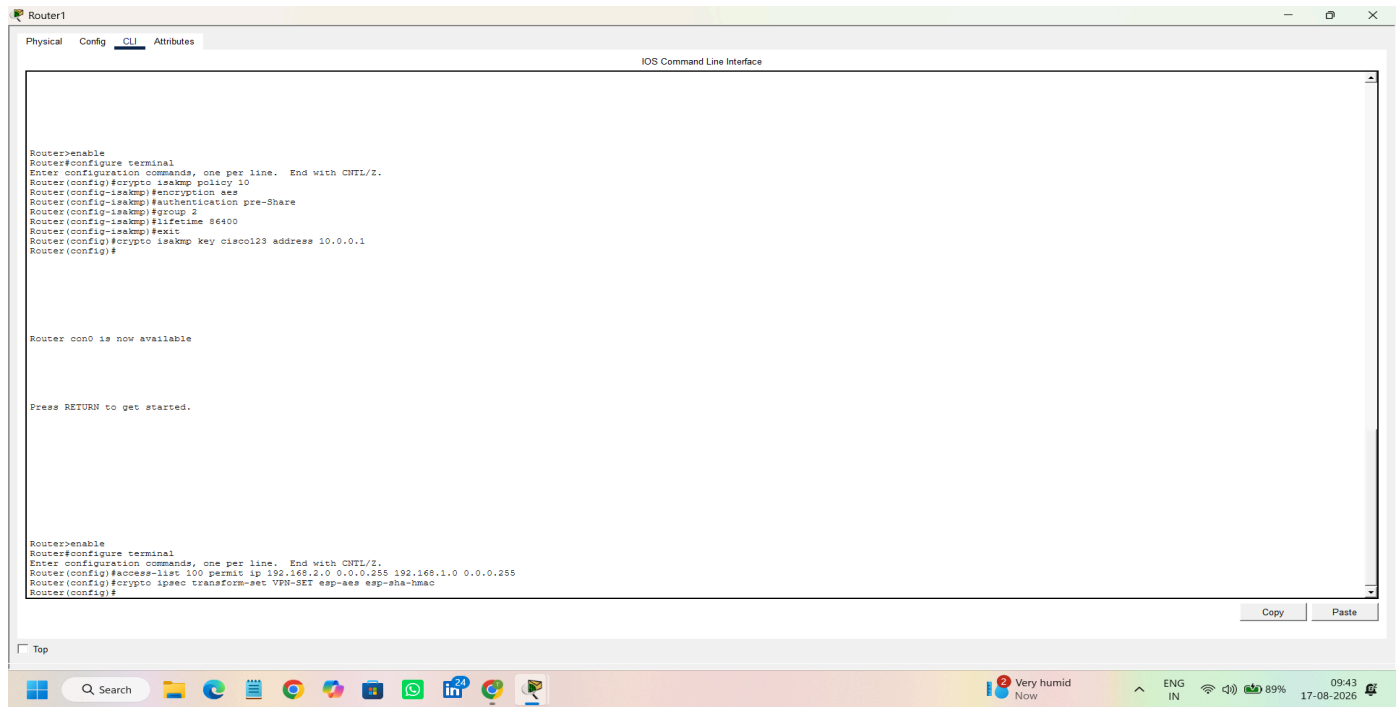
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#authentication pre-Share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#

Router oom0 is now available

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#
```

Step 12:- Configure the Transform Set on Router1



The screenshot shows the Cisco IOS Command Line Interface for Router1. The configuration steps for Step 12 are as follows:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#authentication pre-Share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#

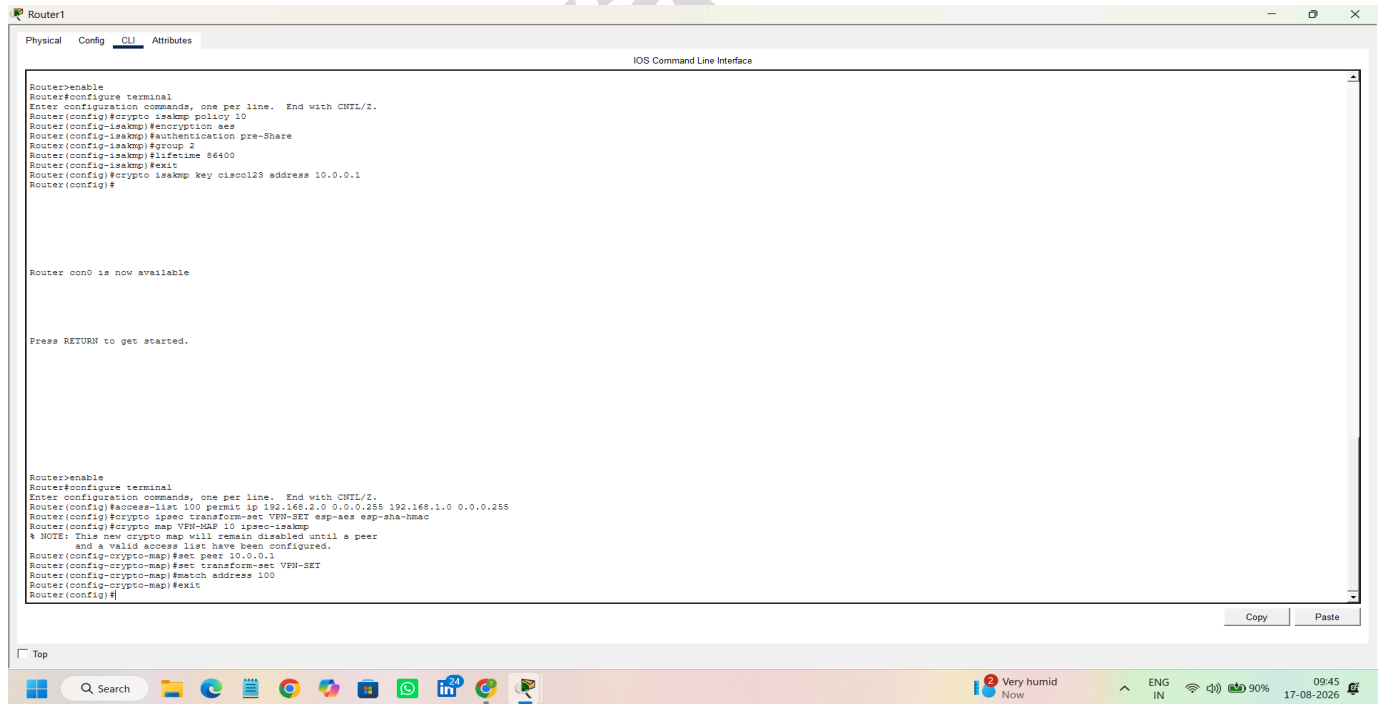
Router con0 is now available.

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#
```

The bottom of the window shows a Windows taskbar with various application icons, a search bar, and system status information including weather (Very humid Now), language (ENG IN), and time (09:43 17-08-2026).

Step 13:- Configure Crypto Map on Router1



The screenshot shows the Cisco IOS Command Line Interface for Router1. The configuration steps for Step 13 are as follows:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#authentication pre-Share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#

Router con0 is now available.

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#
```

The bottom of the window shows a Windows taskbar with various application icons, a search bar, and system status information including weather (Very humid Now), language (ENG IN), and time (09:45 17-08-2026).

SUBJECT: CYBER SECURITY

Step 14:- Apply Crypto Map to Router1

Router1

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Router(config-isaamp)#group 2
Router(config-isaamp)#lifetime 86400
Router(config-isaamp)#exit
Router(config)#crypto isaamp key cisco123 address 10.0.0.1
Router(config)#

Router con0 is now available

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#interface gigabitEthernet 0/1
%Invalid interface type and number
Router(config)#crypto map VPN-MAP
% Incomplete command.
Router(config)#exit
Router#
%SYS-5-CONFIG_I_1: Configured from console by console
```

CopyPaste

Top

Windows Taskbar

29°C Light rain

ENG IN

91%

09:47 17-08-2026

PCO

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Top

29°C Light rain

ENG IN

92%

17-08-2026

09:49

Router

Physical Config CLI Attributes

IOS Command Line Interface

```
% Invalid input detected at '^' marker.
Router(config)#exit
Router#
XVS-5-CONFIG_I: Configured from console by console
Router#interface gigabitEthernet 0/1
% Invalid input detected at '^' marker.
Router#crypto map VPN-IGMP
% Invalid input detected at '^' marker.
Router#exit

Router con0 is now available

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show crypto isakmp sa
% Invalid input detected at '^' marker.
Router(config)#
```

Top

Copy Paste

29°C Light rain ENG IN 92% 09:50 17-08-2026

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```

Router>
+
% Invalid input detected at '' marker.
Router>crypto map VPN-MAP
+
% Invalid input detected at '' marker.
Router>exit

Router con0 is now available

Press RETURN to get started.

Router>enable
Router>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show crypto isakmp aa
+
% Invalid input detected at '' marker.
Router(config)#show crypto ipsec aa
+
% Invalid input detected at '' marker.
Router(config)#show crypto map
+
% Invalid input detected at '' marker.
Router(config)#

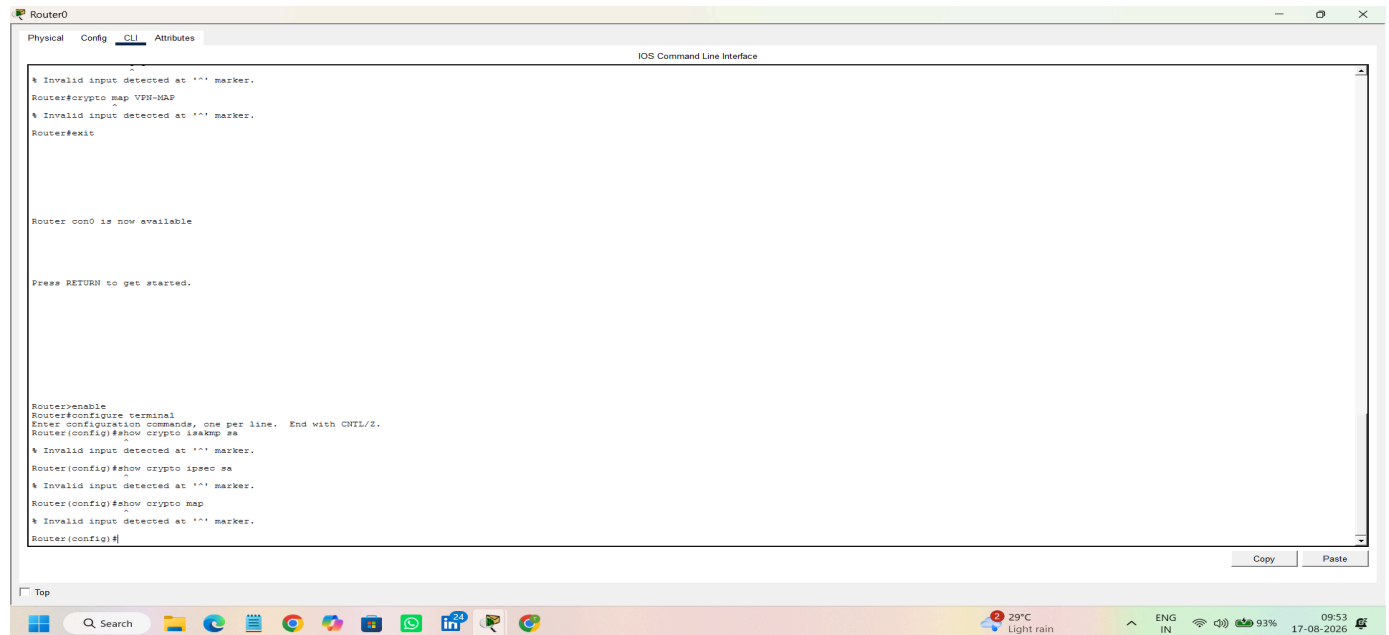
```

Copy Paste

Top

Windows taskbar: Search, File Explorer, Edge, Task View, Chrome, OneDrive, Teams, WhatsApp, LinkedIn, Outlook, 24°C, Light rain, ENG IN, 93%, 09:53, 17-08-2026

Step 18:- Verify the Crypto Map



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#
* Invalid input detected at '^' marker.
Router#crypto map VPN-MAP
* Invalid input detected at '^' marker.
Router#exit

Router con0 is now available

Press RETURN to get started.

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show crypto isakmp sa
* Invalid input detected at '^' marker.
Router(config)#show crypto ipsec sa
* Invalid input detected at '^' marker.
Router(config)#show crypto map
* Invalid input detected at '^' marker.
Router(config)#
```